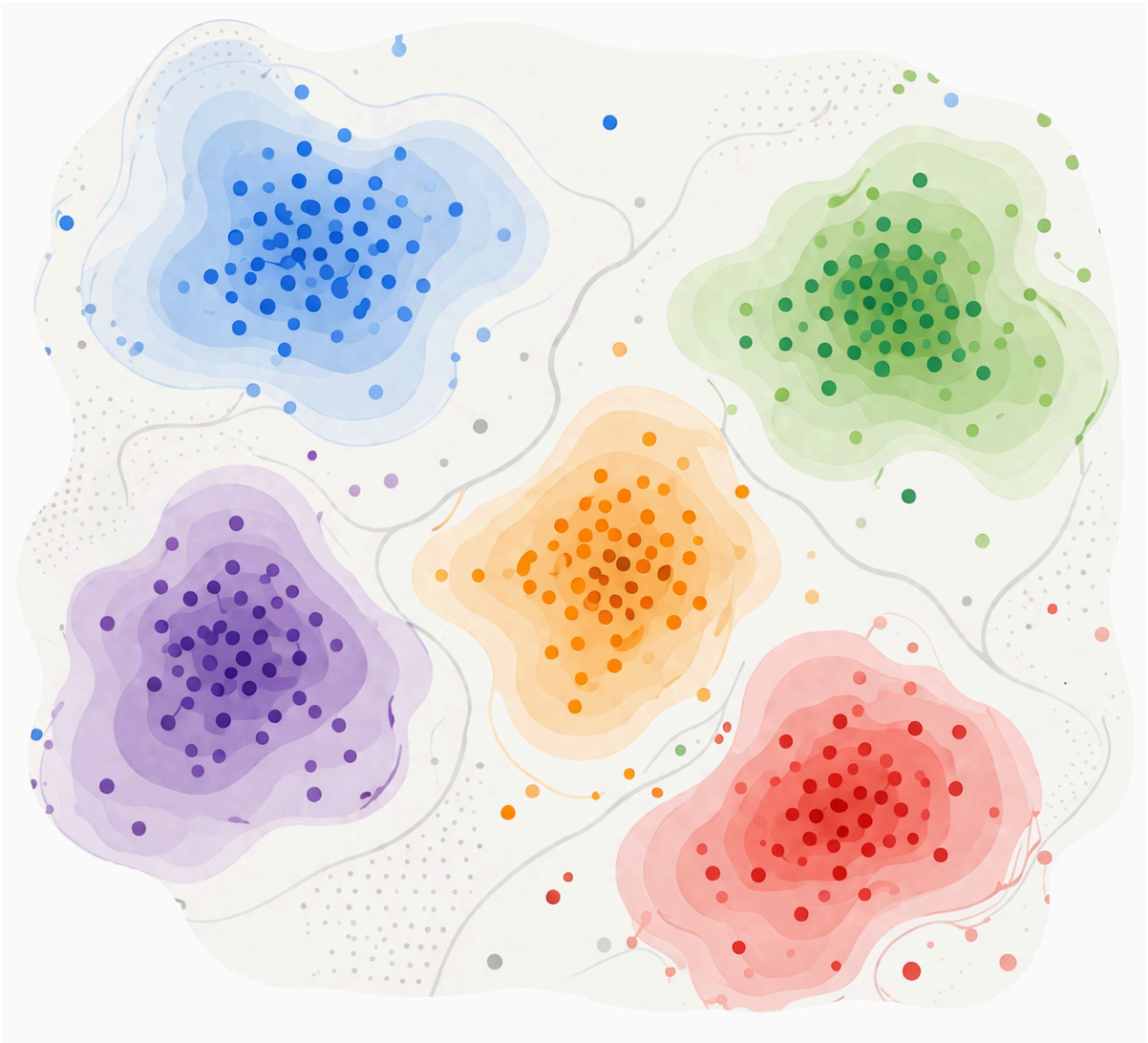
Customer Segmentation Project



In today's competitive market, understanding your customers and tailoring your marketing strategies to meet their specific needs and preferences is critical for business success.

Customer segmentation, the process of dividing a large customer base into smaller groups based on shared characteristics, can help businesses gain valuable insights into their customers and develop targeted marketing strategies that maximize customer engagement and loyalty.

Unsupervised machine learning plays an important role in customer segmentation by uncovering hidden patterns in data without requiring labeled examples. Unlike traditional marketing approaches that rely on predefined customer categories, unsupervised learning allows businesses to discover natural groupings within their customer base based on

purchasing behavior, demographics, browsing history, or other relevant data points. For a data scientist, mastering unsupervised learning algorithms is essential because these techniques enable them to extract valuable insights from data without relying on predefined labels.

For this assignment, you will be provided with two datasets containing information on customer demographics, spending habits, purchasing behavior and historical transitions of the customer. Your task is to perform customer segmentation and identify distinct groups of customers based on their shared characteristics. The two datasets are named - *customer_info (available on the week of April 6)* and *customer_basket (available on the week of April 13).*

The data

Here's a description of the datasets and columns - starting with *customer_info:*

Description:

- **Contains information about each customer, namely demographical and spend behavior for each customer.**

Columns:

- **customer_id**: Identifier of the customer.
- **customer_name**: Name of the customer (contains degree level).
- **customer_gender**: Gender of the customer.
- **customer_birthdate**: Birth date of the customer.
- **kids_home**: Number of kids at home.
- **teens_home**: Number of teens at home.
- **number_complaints**: Number of complaints formally done by the customer.
- **latitude**: Approximate location (<1km range) of the customer's home (Latitude).
- **longitude**: Approximate location (<1km range) of the customer's home (Longitude).
- **distinct_stores_visited**: Number of distinct stores visited by the customer.
- **loyalty_card_number**: Number of the customer loyalty card.
- **lifetime_spend_groceries**: Total value spent by the customer on groceries.
- **lifetime_spend_electronics**: Total value spent by the customer on electronics.
- **lifetime_spend_vegetables**: Total value spent by the customer on vegetables.
- **lifetime_spend_nonalcohol_drinks**: Total value spent by the customer on non-alcoholic drinks.
- **lifetime_spend_alcohol_drinks**: Total value spent by the customer on alcoholic drinks.
- **lifetime_spend_meat**: Total value spent by the customer on meat.
- **lifetime_spend_fish**: Total value spent by the customer on fish.
- **lifetime_spend_hygiene**: Total value spent by the customer on hygiene.
- **lifetime_spend_petfood**: Total value spent by the customer on pet food.
- **lifetime_spend_videogames**: Total value spent by the customer on video games.
- **lifetime_total_distinct_products**: Number of distinct products bought by the customer (lifetime).
- **year_first_transaction**: Year of the first transaction of the customer.
- **percentage_of_products_bought_promotion**: Percentage of products that were bought with some promotion.
- **typical_hour**: Typical hour when the customer visits the store

Regarding *customer_basket,* that file contains 100.000 **randomly sampled baskets from our customers:**

Description:

- Contains information about customers' different baskets bought at the shop.
- Each line consists of the customer_id, transaction_id and products in list format.

Columns:

- **invoice_id:** Identifier of the transaction.
- **list_of_goods:** Products bought in the transaction in list format.
- **customer_id:** Identifier of the customer. Can be used to connect this dataset with the *customer_info* one.

The project

In this project, you will be asked to:

- Identify relevant customer segments: Using statistical and machine learning techniques, you will need to identify meaningful segments within the customer base that share similar characteristics. These clusters can then be used to develop targeted marketing strategies

- Analyze customer behavior: Once you have identified customer segments, you will need to analyze their behavior to gain insights into their motivations, preferences, and needs. This may involve analyzing their purchasing patterns, loyalty card usage, complaint history, and other relevant data.

- Develop targeted marketing strategies: Using the customer_basket dataset, you will need to develop targeted marketing strategies that appeal to each customer segment's unique needs and preferences. This may involve developing personalized promotions, creating targeted advertising campaigns, or tailoring product offerings to meet specific customer demands.

Overall, customer segmentation is a powerful tool that can help businesses gain a competitive edge by understanding their customers better and developing targeted marketing strategies that maximize customer engagement and loyalty. By completing this assignment, you will gain hands-on experience with unsupervised learning and develop valuable skills that can help you succeed in a wide range of data science roles.

Deliverables

You will be asked to deliver a repository with code and explanation of the solution. Deliver code and report using the repo by inviting github user *ivopbernardo* to your repo.

Clean, "DRY" (Don't repeat yourself) and clear code will earn extra points in the submission. **You should abstract most code (using .py files) outside of notebooks so that the code is versionable in the git repository. Unless you state otherwise, the last commit before June 9th at 23:59:59 will be considered. If you want another later commit to be evaluated, let ibernardo@novaims.unl.pt know and you will get the penalty for every day of delay.**

In addition, you will have to submit an executive report with the detail of each segment and suggested campaigns that you would do for each segment - **be creative on the campaign suggestion** - here are some examples:

- Get an extra 20% on Fish if you buy meat.
- 50% promotion on the entire basket just for you!
- Buy 1 Video Game, get 1 free.

To come up with the promotions, you can play around with association rules on the *customer_basket* dataset and relate those baskets with the customers on the *customer_info* one.

A **suggested** index of the report is the following:

- 1) Executive Summary
- 2) Exploratory Data Analysis and Pre-Processing
- 3) Customer Segmentation and Clustering
- 4) Targeted Promotion
- 5) Conclusion and recommendations

As an alternative to the report, you may also submit a working application, instead of a PDF, deliver a runnable app (e.g. a Streamlit app) that presents the segmentation results, cluster profiles, and campaign suggestions interactively. The app will be executed and evaluated directly, so make sure it runs without errors from a clean environment. If you choose this option, include a short README with the instructions to run it.

No code should be presented during the report or app - code should be reserved for the Notebook(s) and py files. Additionally, you should deliver a csv file with each customer_id and its proposed cluster. Make sure that every customer is included in the final file.

Evaluation

The project will be evaluated based on code quality, cleanness and overall quality of the segmentation solution. There is an “ideal solution” and your clustering solution will be considered against that “ground truth” solution.

In terms of evaluation parameters, the following will be considered:

- Technical quality of the Segmentation: **5 Points**
- Git Usage: **3 Points**
- Quality of the Report / Application: **4 Points**
- Visuals and Explanation: **3 Points**
- Interpretation of the Clusters: **2 Points**
- Feasibility of Promotions vs. Cluster Interpretation: **3 Points**
- Extra: Code quality: **+2**